

Multiple Choice (10 marks)

1. In Python 3, which of the following function sets the integer starting value used in generating random numbers?
 - (a) choice(seq)
 - (b) randrange ([start,] stop [,step])
 - (c) random()
 - (d) seed([x])
2. In Python 3, what is the following function returns item from the list with max value?
 - (a) cmp(list)
 - (b) len(list)
 - (c) max(list)
 - (d) min(list)
3. Which is the smallest asymptotically?
 - (a) $O(1)$
 - (b) $O(n)$
 - (c) $O(n^2)$
 - (d) $O(\log n)$
4. In Python 3, which of the following operator in python performs exponential (power) calculation on operands?
 - (a) **
 - (b) //
 - (c) is
 - (d) not in
5. Which types of functions grow the slowest?
 - (a) $O(\log e^N)$
 - (b) $O(\log N)$
 - (c) $O(\log \log N)$
 - (d) $O(N)$

True / False (10 marks)

Answer True or False

6. True or False: Authentication is compromised when a user logs into a system with stolen credentials.
7. True or False: The sum of the list $l = [1, 2, 3, 4]$ in Python3 is 8.
8. True or False: In Python 3, the operator `'/'` is used for floor division.
9. True or False: The hexadecimal representation of the decimal number 231 is $E7_{16}$.
10. True or False: The function $O(N^{1/N})$ grows the slowest compared to other common growth rates as N increases.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how the RGB color model represents colors through its three components: red, green, and blue. Analyze how many bits are necessary to represent a single color and explain your reasoning.
12. Discuss the concept of fault tolerance in Internet routing. How does it ensure data transmission despite the failure of some connections? Provide examples to support your analysis.

End of Paper